Z-Scores & Linear Transformations

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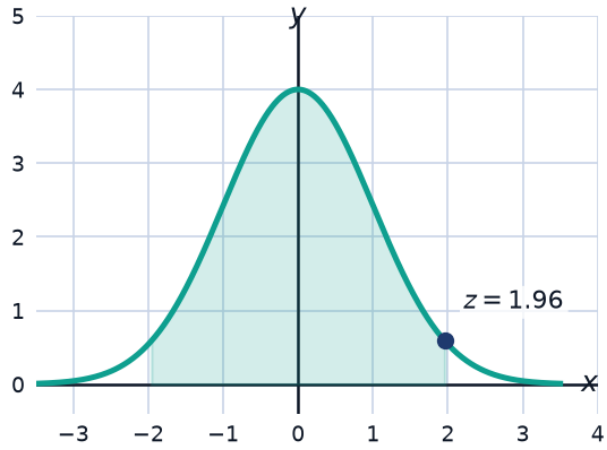
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Computing z-scores

A test has mean 70 and standard deviation 5. Find the z-score for a student who scored 82.

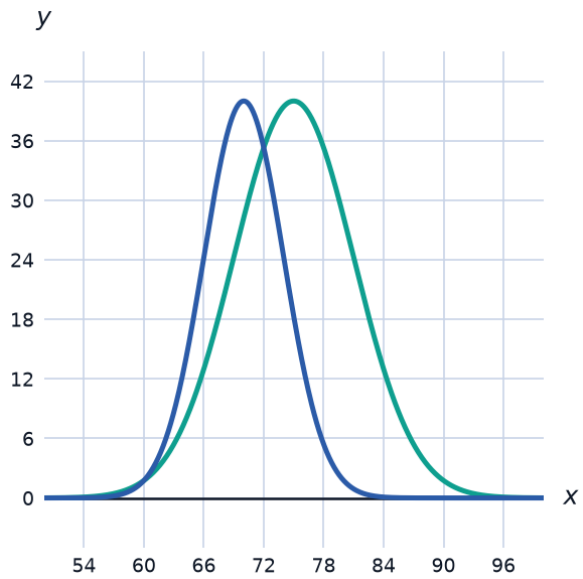
2 [Foundational] • Target: Computing z-scores

Explain in words what a z-score of -1.5 means, in the context of a data value's position relative to its distribution.



3 [Foundational] • Target: Comparing values across different distributions

On Test A (mean 75, standard deviation 6), a student scores 87. On Test B (mean 70, standard deviation 4), a different student scores 77. Compare the two students' relative performance using z-scores.



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4 [Foundational] • Target: Effect of adding a constant to every data value

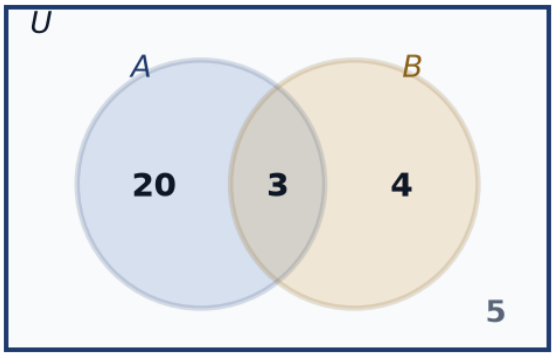
A data set has mean 50 and standard deviation 8. If 10 is added to every value, find the new mean and new standard deviation.

5 [Foundational] • Target: Effect of adding a constant to every data value

Explain, without doing a full calculation, why adding a constant to every data value does not change the standard deviation.

6 [Foundational] • Target: Effect of multiplying every data value by a constant

A data set has mean 20 and standard deviation 3. If every value is multiplied by 4, find the new mean and new standard deviation.



7 [Foundational] • Target: Effect of multiplying every data value by a constant

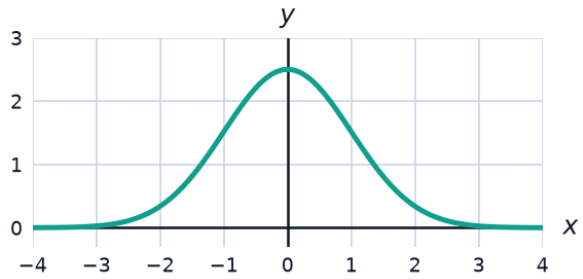
A distance measured in miles has mean 10 and standard deviation 2. Convert both to kilometers (multiply by 1.609), and find the new mean and standard deviation, to two decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Visualizing a standardized (z-score) distribution

The graph shows the standard normal distribution, which has mean 0 and standard deviation 1 — the distribution that results after standardizing any normal data set using z-scores.



a State the mean and standard deviation shown by this curve.

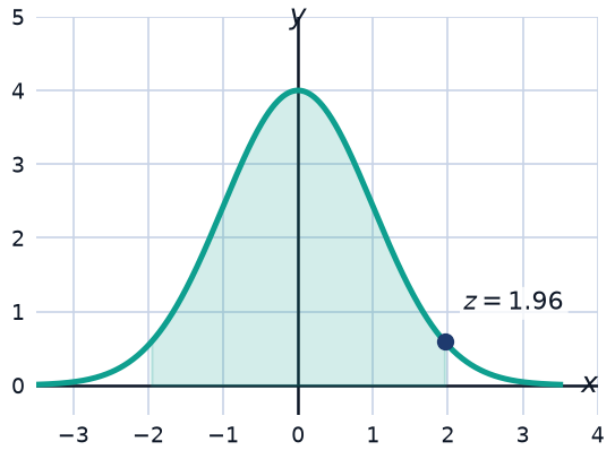
b Explain why standardizing a data set using $z = \frac{x - \mu}{\sigma}$ always produces a distribution with mean 0 and standard deviation 1, regardless of the original data's mean and standard deviation.

9 [Intermediate] • Target: Applying linear transformations in context

A factory's daily output (in units) has mean 500 and standard deviation 30. Management adds a fixed bonus batch of 50 units to every day's count, then reports output in dozens (divide by 12). Find the new mean and standard deviation, to two decimal places.

10 [Intermediate] • Target: Finding a raw value from a z-score

A distribution has mean $\mu = 64$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.8$.



11 [Intermediate] • Target: Finding a raw value from a z-score

A distribution has mean $\mu = 200$ and standard deviation $\sigma = 15$. If a data value has $z = 2.2$, find the data value.

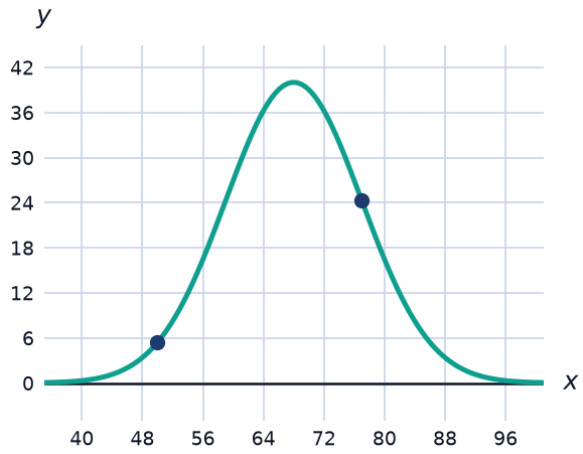
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12 [Intermediate] • Target: Combined shift-and-scale transformations

A teacher curves test scores (mean 65, standard deviation 10) by first adding 5 points to every score, then multiplying the result by 1.1. Find the new mean and new standard deviation, to two decimal places.

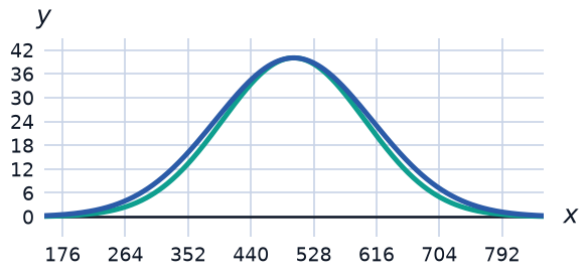
13 [Intermediate] • Target: Combined shift-and-scale transformations

A weight distribution has mean $\mu = 68$ kg and standard deviation $\sigma = 9$ kg. Find the z-scores for weights of 77 kg and 50 kg.



14 [Intermediate] • Target: Combined shift-and-scale transformations

A national math exam has mean 500 and standard deviation 100, while a national verbal exam has mean 500 and standard deviation 110. A student scores 640 on the math exam and 625 on the verbal exam. Even though the verbal raw score is lower, use z-scores to determine which exam the student performed better on relative to other test-takers, and explain your reasoning.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

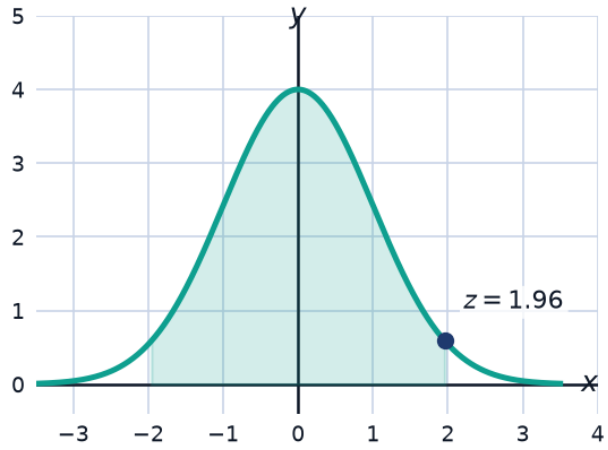
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15 **[Challenging]** • **Target: Combined shift-and-scale transformations** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A shipping company records package weights with mean 18 kg and standard deviation 2.5 kg. The company then converts every measurement from kilograms to pounds by multiplying by 2.20462, and separately subtracts 1 kg from every package to account for packaging weight before converting. Describe, step by step, how each operation (the subtraction, then the multiplication) affects the mean and standard deviation.

16 **[Challenging]** • **Target: Effect of multiplying every data value by a constant** • **Cross-Lesson: 3.03 The Normal Distribution**

A manufacturing company produces electronic components. They are investigating potential relationships between the type of raw material used and the number of defects per batch. After collecting data for 100 batches, they create a two-way table summarizing the findings. The table shows that 15 batches using Material A had no defects, 20 batches using Material A had 1-2 defects, and 10 batches using Material A had more than 2 defects. For Material B, 25 batches had no defects, 28 batches had 1-2 defects, and 22 batches had more than 2 defects. If the company decides to perform a chi-square test for homogeneity to compare the defect rates across the two materials, what is the degrees of freedom for this test?



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17 **[Challenging]** • **Target: Visualizing a standardized (z-score) distribution** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A research team is investigating the relationship between socioeconomic status (SES) and preferred mode of transportation for commuting to work in a large metropolitan area. They conduct a survey and categorize SES into three levels: Low, Medium, and High. The preferred mode of transportation is categorized as Public Transit, Private Vehicle, or Bicycle/Walking. After collecting data from a sample of $n = 500$ individuals, they decide to perform a chi-square test for independence. The preliminary analysis suggests that the expected counts for several cells are small. To ensure the validity of the chi-square approximation, the research team decides to merge categories. They consolidate 'Bicycle/Walking' into 'Public Transit' to form a new category 'Active/Public Transit'. If the original analysis involved 3 levels of SES and 3 modes of transportation, what is the minimum number of observations required in each cell of the *original* contingency table for the chi-square test to be considered reasonably reliable, assuming equal distribution across all cells?

18 **[Challenging]** • **Target: Applying linear transformations in context** • **Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)?

19 **[Challenging]** • **Target: Finding a raw value from a z-score** • **Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A political scientist is analyzing voter preferences for two candidates, Candidate A and Candidate B, across different age groups. The data is summarized in the two-way table below. The total number of surveyed individuals is $N = 500$. The total number of individuals in the 'Young Adults' row is $R_1 = 120$, and the total number of individuals who prefer Candidate A (Column 1) is $C_1 = 200$. If the political scientist were to conduct a chi-square test of independence on this data, what would be the expected count for the cell representing 'Young Adults' who prefer 'Candidate A'?

20 **[Challenging]** • **Target: Combined shift-and-scale transformations** • **Cross-Lesson: 1.01 Measures of Center & Spread**

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Computing z-scores

A distribution has mean $\mu = 51$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 41$.

a. -0.40

b. 0.40

c. -0.90

d. 0.10

22 [Foundational] • Target: Comparing values across different distributions

A distribution has mean $\mu = 30$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = 66$.

a. 18.50

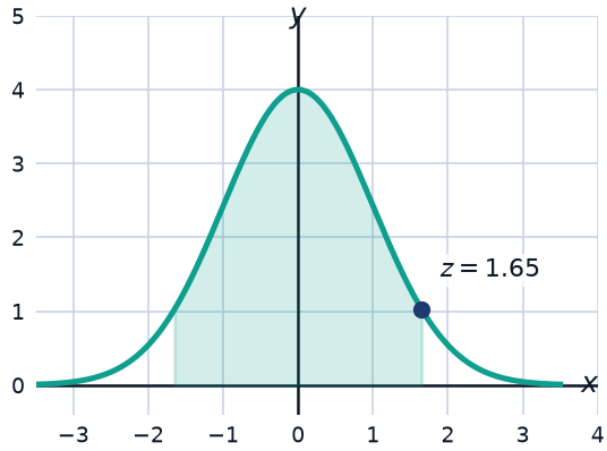
b. -18.00

c. 18.00

d. 17.50

23 [Foundational] • Target: Effect of adding a constant to every data value

A distribution has mean $\mu = 50$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = -5$.



a. -27.50

b. -28.00

c. -27.00

d. 27.50

24 [Foundational] • Target: Effect of multiplying every data value by a constant

A distribution has mean $\mu = 77$ and standard deviation $\sigma = 25$. Find the z-score for a value of $x = 50$.

a. -0.58

b. 1.08

c. -1.58

d. -1.08

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25 [Foundational] • Target: Visualizing a standardized (z-score) distribution

A distribution has mean $\mu = 78$ and standard deviation $\sigma = 8$. Find the z-score for a value of $x = 19$.

a. 7.38

b. -7.38

c. -7.88

d. -6.88

26 [Foundational] • Target: Applying linear transformations in context

A distribution has mean $\mu = 80$ and standard deviation $\sigma = 20$. Find the z-score for a value of $x = 119$.

a. 1.95

b. 1.45

c. 2.45

d. -1.95

27 [Foundational] • Target: Finding a raw value from a z-score

A distribution has mean $\mu = 69$ and standard deviation $\sigma = 2$. Find the z-score for a value of $x = 44$.

a. -12.00

b. 12.50

c. -12.50

d. -13.00

28 [Foundational] • Target: Combined shift-and-scale transformations

A distribution has mean $\mu = 85$ and standard deviation $\sigma = 25$. Find the data value corresponding to a z-score of $z = -1.4$.

a. 49

b. 51

c. 120

d. 50

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Calculating and Interpreting Z-Scores

A distribution has mean $\mu = 76$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = 1.3$.

a. 101

b. 103

c. 102

d. 50

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30 [Intermediate] • Target: Linear Transformations - Adding a Constant

A distribution has mean $\mu = 180$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = -2.2$.

a. 224

b. 137

c. 135

d. 136

31 [Intermediate] • Target: Linear Transformations - Multiplying by a Constant

A distribution has mean $\mu = 123$ and standard deviation $\sigma = 25$. Find the data value corresponding to a z-score of $z = 0.4$.

a. 113

b. 132

c. 133

d. 134

32 [Intermediate] • Target: Standardizing an Entire Distribution

A distribution has mean $\mu = 50$ and standard deviation $\sigma = 20$. Find the data value corresponding to a z-score of $z = 2.4$.

a. 98

b. 97

c. 99

d. 2

33 [Intermediate] • Target: Computing z-scores

A distribution has mean $\mu = 57$ and standard deviation $\sigma = 10$. Find the data value corresponding to a z-score of $z = -1.9$.

a. 37

b. 38

c. 39

d. 76

34 [Intermediate] • Target: Comparing values across different distributions

A distribution has mean $\mu = 111$ and standard deviation $\sigma = 8$. Find the data value corresponding to a z-score of $z = 0.5$.

a. 115

b. 107

c. 114

d. 116

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35 **[Intermediate]** • **Target: Effect of adding a constant to every data value**

On Test A (mean 52, standard deviation 4), a student scores 53. On Test B (mean 54, standard deviation 5), a different student scores 61. Using z-scores, which student performed better relative to their own test's distribution?

a. Test A

b. Both are exactly the same

c. Cannot be determined without more information

d. Test B

36 **[Intermediate]** • **Target: Effect of multiplying every data value by a constant**

On Test A (mean 90, standard deviation 10), a student scores 94. On Test B (mean 76, standard deviation 4), a different student scores 86. Using z-scores, which student performed better relative to their own test's distribution?

a. Both are exactly the same

b. Test A

c. Test B

d. Cannot be determined without more information

37 **[Intermediate]** • **Target: Visualizing a standardized (z-score) distribution**

A data set has mean 40 and standard deviation 18. First, 5 is added to every value; then every value is multiplied by 2. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(90, 18)$

b. $(80, 36)$

c. $(91, 36)$

d. $(90, 36)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Applying linear transformations in context • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A data set has mean 40 and standard deviation 3. First, -7 is added to every value; then every value is multiplied by 4. Find the new mean and standard deviation as an ordered pair (mean, SD).

a. $(132, 12)$

b. $(133, 12)$

c. $(160, 12)$

d. $(132, 3)$

39 [Challenging] • Target: Finding a raw value from a z-score • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A distribution has mean $\mu = 41$ and standard deviation $\sigma = 8$. Find the z-scores for values of 52 and 31, as an ordered pair (z_{52}, z_{31}), to two decimal places.

a. $(1.38, -1.25)$

b. $(1.68, -1.25)$

c. $(-1.25, 1.38)$

d. $(1.38, -1.55)$

40 [Challenging] • Target: Combined shift-and-scale transformations • Cross-Lesson: 1.01 Measures of Center & Spread

A distribution has mean $\mu = 53$ and standard deviation $\sigma = 5$. Find the z-scores for values of 74 and 47, as an ordered pair (z_{74}, z_{47}), to two decimal places.

a. $(4.50, -1.20)$

b. $(4.20, -1.50)$

c. $(-1.20, 4.20)$

d. $(4.20, -1.20)$

41 [Challenging] • Target: Calculating and Interpreting Z-Scores • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A student scores 82 on an exam where the class mean is 74 and the standard deviation is 5. What is the student's standardized z-score?

a. 1.60

b. 1.20

c. 0.80

d. -1.60

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42 [Challenging] • Target: Calculating and Interpreting Z-Scores • Cross-Lesson: 2.01 Sampling Methods & Bias

On Test 1, Alice scored 88 (mean 80, SD 4). On Test 2, she scored 92 (mean 85, SD 5). On which test did she perform relatively better compared to peers?

a. Test 1 ($z_1 = 2.0 > z_2 = 1.4$)

b. Test 2 ($z_2 = 1.4 > z_1 = 1.0$)

c. Both performances were identical

d. Test 2 because her raw score was 92 vs 88

43 [Challenging] • Target: Linear Transformations - Adding a Constant • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A teacher curves a test by adding 8 points to all student scores. How does this affect the shape of the score distribution?

a. The shape remains exactly the same

b. The distribution becomes more symmetric

c. The distribution becomes skewed to the right

d. The spread increases

44 [Challenging] • Target: Linear Transformations - Multiplying by a Constant • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

If all observations in a dataset with variance 16 are multiplied by -3, what is the new variance?

a. 144

b. -48

c. 48

d. 12

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45 [Challenging] • Target: Standardizing an Entire Distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

If an entire non-normal distribution with mean $\mu = 40$ and $\text{SD} = 6$ is converted to z-scores ($z = \frac{x - 40}{6}$), what are the mean and standard deviation of the standardized values?

a. Mean $= 0$, Standard Deviation $= 1$

b. Mean $= 40$, Standard Deviation $= 1$

c. Mean $= 0$, Standard Deviation $= 6$

d. Mean $= 1$, Standard Deviation $= 0$

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